

Preventable Death in Forward Surgical Care: Casualty Surge and Contested Evacuation in Large-Scale Combat

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Executive Summary

Two decades of counterinsurgency under owned skies drove combat case-fatality to historic lows (Remondelli et al., 2023). Large-scale combat against a peer removes the two conditions behind those rates: casualties arrive in surges, and contested airspace lengthens evacuation from the golden hour, the first hour after wounding when reaching surgery most improves survival (Kotwal et al., 2016), to hours or days. Pairing Army casualty-planning doctrine with a simulation of the forward resuscitative surgical detachment (FRSD), the two-table surgical unit nearest the fight, we count the deaths caused by slow evacuation or a wait for an open table, not by the wound itself. Varying the casualty surge, air superiority, and surgical posture, we find that a single FRSD sized to current doctrine loses about one in five operative casualties under permissive air; that losing air superiority more than doubles that toll, which adding surgical capacity does not; and that air superiority decides the better posture, with surgery consolidated while the air is owned and distributed once it is contested. This answers the Office of The Surgeon General’s question on medical operations in large-scale combat (Strategic Studies Institute, 2025, p. 98) and a companion question on the strategic cost of contested evacuation (Cunningham, 2020, p. 29).

1. Introduction

For two decades, counterinsurgency moved the wounded to surgery quickly and through skies the United States controlled, and combat case fatality fell to historic lows (Eastridge et al., 2012; Katzman et al., 2025). A peer war removes both conditions. Casualties arrive in surges, and contested airspace lengthens evacuation from the golden hour to hours or days (Lawry et al., 2025). At the FRSD, casualties can arrive faster than they can be treated, and some die of their wounds while they wait. Those are preventable deaths, the ones faster or better-arranged care would have saved, and they are the only share force design can reduce. We answer the Surgeon General’s question on medical operations in large-scale combat (Strategic Studies Institute, 2025, p. 98) and a companion question on contested evacuation (Cunningham, 2020, p. 29) by pairing Army casualty-planning doctrine with a simulation of the front line as a queue where the FRSD provides the surgery and waiting can be fatal, then varying the inputs we trust least to measure how that toll responds.

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2. Approach

Doctrinally, the FRSD has two operating tables and handles about ten cases a day (HQDA, 2020a). Army planning factors put the offensive casualty rate at 3.54 to 15.60 wounded per 1,000 per day, from a lower-intensity fight to the most intense, loading a single FRSD from 21 to 94 percent of its capacity (Table E.1) (HQDA, 2020b). Against that demand, we run the queue as a discrete-event simulation in two mirrored worlds, the same casualties and wounds in each. One keeps realistic evacuation and surgery times, with the delays and table-waits of contested combat; the other moves and operates on everyone instantly. A casualty who dies in the realistic world but lives in the instant one was killed by the delay and counts as preventable; one who dies even in the instant world could not be saved by faster care, and does not. We run the simulation across a variety of plausible assumptions and scenarios. The full model, with its accompanying assumptions and parameters, is in the appendices. The online companion runs the same simulation interactively, where the reader can turn the inputs and watch preventable death respond, alongside the fuller set of results and methods.

3. Findings

Optimal posture depends on the level of air superiority. A fixed surgical force can be consolidated into a few large hospitals to the rear or distributed across many small teams near the front. Consolidation pools demand and works down surges faster but lengthens the trip to a table; distribution shortens that trip but leaves each node small and quicker to swamp in a surge. Consolidation wins under permissive and degraded air, distribution wins under contested air, with the transition point near the air-denied end of the range (Figure E.1). That point is the model’s output, not a number we chose, and wherever air superiority is in doubt there is a level past which distribution wins, precisely the scenario large-scale combat against a peer risks generating.

Distributing forward helps only a little. At its best it recovers about 40 per 1,000 of the roughly 280 that losing air superiority costs, and that gain comes from the shorter evacuation leg alone (Appendix E).

Evacuation is the dominant lever. Varying only air superiority, preventable death more than doubles, from about 219 to about 499 per 1,000 operative casualties. This is the strategic cost of contested evacuation, and the largest structural risk the model finds. Doubling the operating tables cuts that toll by at most about 14 per 1,000, far less than the roughly 280 that lost air superiority adds (Figure E.1).

Arrivals within the daily-average estimate overrun the FRSD when they surge. Even at the worst doctrinal load, the FRSD’s workload reaches only 94% of its capacity (Table E.1). The same daily caseload arriving steadily kills about 22 per 1,000, but bunched into a surge it makes casualties wait at once and the toll nearly doubles, to about 38 (Appendix E).

4. What it means

These are results for the Medical Corps to examine. The general result is that the better posture switches with air. Wherever air superiority is in doubt, there is a point past which the distributed posture wins, so for a future operation under uncertain air, a more forward FRSD, one that holds blood and pushes high-payoff triage toward the front, may genuinely make more sense. Behind it, evacuation outweighs added capacity as the lever on preventable death, and the surge, not the daily average, sets the requirement. The reported levels are conservative, set by a clinical survival curve we could not pin.

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Appendix A. Queueing formulation and the closed form

The single node is M/M/c+M (Erlang-A): Poisson arrivals, c exponential servers, exponential patience. The stationary birth-death law gives the abandonment probability P_{ab} (the died-of-wounds fraction), the wait probability, and the queue length in closed form. As $\rho = \lambda/(c\mu)$ passes one the abandoning fraction approaches the fluid limit $(1 - 1/\rho)^+$, and the transition sharpens with server count as $1/\sqrt{c}$ (Garnett et al., 2002; van Leeuwen et al., 2019). A general-patience extension (M/M/c+G) carries a non-exponential survival clock; it is exact when patience is exponential and a mean-matched approximation otherwise, biased fifteen to twenty-five percent against the simulation, so the simulation carries the survival-clock shape. The closed form matches an independent continuous-time Markov chain solve to 3.7×10^{-12} across the grid, and a hand check at $c=1$, $\lambda=\mu=\theta$ returns exactly $1/e$. The simulation, reduced to these assumptions, reproduces exact Erlang-A on 52 of 54 grid cells within Monte-Carlo error.

Appendix B. The posture model

A division slice of about 18,000 carries a fixed total of eight surgical tables divided across K nodes, with $c = 8/K$ tables per node. Because the total is fixed, per-node offered load is identical across postures ($\rho = 0.59$ at the high-intensity load), so the comparison isolates arrangement from load. The ground leg shortens with node count along $L_{\min} + (L_{\max} - L_{\min}) K^{-\beta}$, $\beta = 0.5$, $L_{\min} = 60$ and $L_{\max} = 480$ minutes, endpoints pinned to the evacuation literature. Air contest sets the air-evacuation fraction (0.90/0.50/0.10) and the ground-leg congestion multiplier ($\times 1/ \times 2/ \times 5$) for permissive, degraded, and contested air.

Appendix C. The estimand: paired counterfactual

Each casualty draws one survival clock and one intrinsic-survivability outcome, consumed by both arms. The ideal arm is instant evacuation and immediate surgery, so any salvageable casualty survives it; a casualty drawn intrinsically non-survivable dies in both arms and is not preventable. The per-class non-survivable fractions (0.01/0.05/0.20/0.75 by severity) are converted from the ideal-care mortality ladder of Eastridge et al. (2012, p. S435). Preventable, non-preventable, and saved counts are reconciled by conservation invariants checked independently of the engine’s own counters.

Appendix D. Inputs and provenance

Sixteen loaded inputs, each tagged measured, derived, assumed, or doctrine. Demand: severity mix (Hall et al., 2023, measured), offensive WIA rates (HQDA, 2020b, doctrine), per-team catchment (swept band, assumed). Capacity and service: tables and ceiling (HQDA, 2020a, doctrine), class service means (Hall et al., 2023, measured), service variability (derived). Survival clock: a Weibull family fit to the dynamic-mortality curves (Mitchell et al., 2004, measured), with the one-hour survival level anchored to the high, medium, and low bins, the Weibull shape an analyst re-fit, and the scale (the level) the primary swept axis. Intrinsic mortality floor (Eastridge et al., 2012, derived). No single source feeds both an input and the conclusion that input would drive.

Table E.1: Doctrinal demand and capacity. Offensive infantry WIA rates (HQDA, 2020b) bracket the high-intensity range; the per-node load is the offered load a single doctrinal team carries at each rate.

Operating point	WIA / 1,000 / day	Source (ATP 4-02.55)	Per-node load ρ
Lower-intensity offensive	3.54	Table D-4h, p. D-10	0.21
High-intensity offensive	9.74	Table D-4c, p. D-7	0.59
Peak offensive	15.60	Table D-4d, p. D-8	0.94

Capacity: 2 tables, ~ 10 cases / 24h, ~ 135 min / case (HQDA, 2020a). Full provenance in Appendix D.

Appendix E. Full results and the stress battery

By doctrinal rate the single team runs at twenty-one, fifty-nine, and ninety-four percent of capacity across the lower-intensity, high-intensity, and peak offensive rates; preventable death among operative casualties is already about twenty-two percent under permissive air at these rates. Holding the daily caseload fixed and switching from steady arrivals to a surge raises died-of-wounds from about 2 to 23 per 1,000 at the lowest rate, 10 to 31 at the high-intensity rate, and 22 to 38 at the peak. Sweeping air contest g from permissive ($g=0$) to contested ($g=1$), with air-evacuation availability $0.9 - 0.8g$ and ground congestion $1 + 4g$, the consolidated and distributed curves meet near $g \approx 0.63$ (Figure E.1). The stress battery (Table E.3) builds each test to overturn the posture finding; the finding holds in all but the last, the attrition level at which it flips.

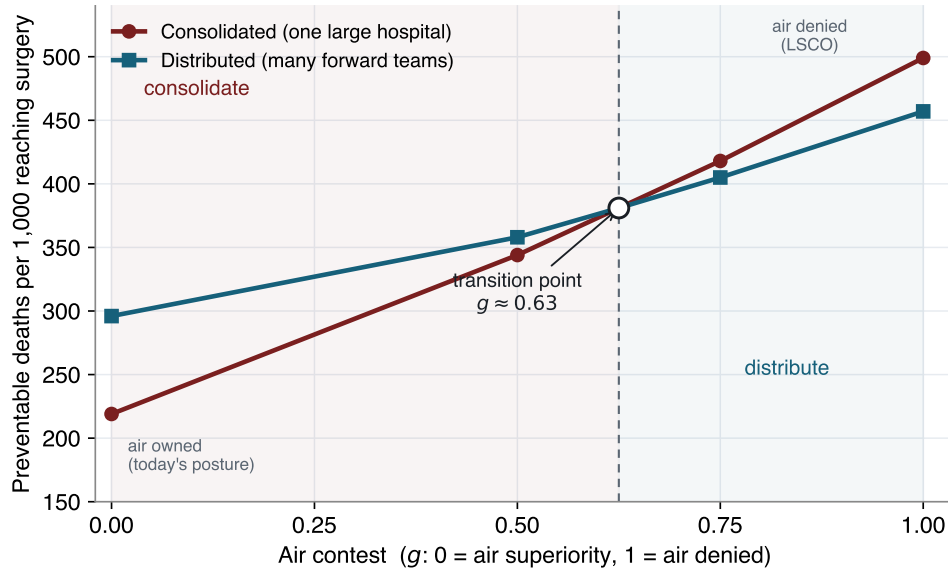


Figure E.1: The posture transition at the high-intensity load. Preventable deaths per 1,000 operative casualties as air control degrades from owned ($g=0$) to denied ($g=1$). Consolidation is more efficient and wins while air is owned; distribution is faster and wins once air is largely denied. The two curves meet at $g \approx 0.63$.

Table E.2: Posture surface: preventable deaths per 1,000 operative casualties at the high-intensity load. The ordering holds at all three doctrinal rates.

Air picture	Consolidated ($K=1$)	Distributed ($K=8$)	Better posture
Permissive	219	296	consolidated (by 77)
Degraded	331	347	consolidated (by 16)
Contested	499	457	distributed (by 42)

Table E.3: Anti-tautology stress battery. Each test is built to overturn the posture finding.

Test	Result
No proximity benefit ($\beta = 0$)	distribution loses at every air level; the contested-air win is entirely the modeled shorter leg
Permissive air	consolidation wins at every load; the model is not pre-committed
Worst-case fragmentation (4 tables, single-table nodes)	distribution still wins contested air, by 28 per 1,000
Full sourced leg range ($L_{\max} \in \{420, 480, 540\}$, congestion $\{4, 5, 6\}$)	distribution wins all nine cells, by 38 to 49
Deep strike on forward nodes	advantage flips to consolidation only near twenty percent per-node attrition
LSCO-heavy severity mix	distribution still wins, margin shrinks from 42 to 27 as intrinsic mortality caps the benefit